NYC Taxi Data Analysis: Comparing January 2024 & January 2025

Team Members:

- **Karthik Vinay Kumar Adari**
- **Yojitha Naga Vaibhavi Uppala**
- **Anuhya Vaagita**

Business Problem Statement

Objective:

Analyze NYC taxi data to understand daily demand, hotspot locations, and fare trends.

Why It Matters:

Helps improve resource allocation, optimize pricing strategies, and informs urban planning decisions.

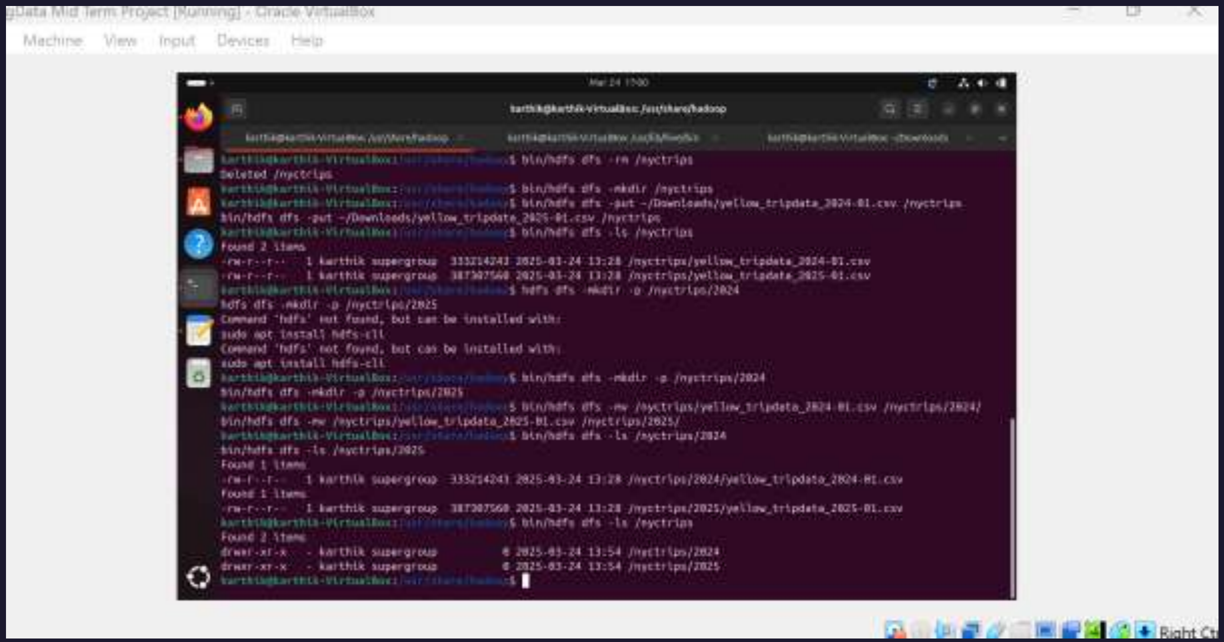
Key Questions:

How do daily trip volumes vary?

Which pickup locations show the highest demand?

How do average fares change over time?

- **Data Overview**

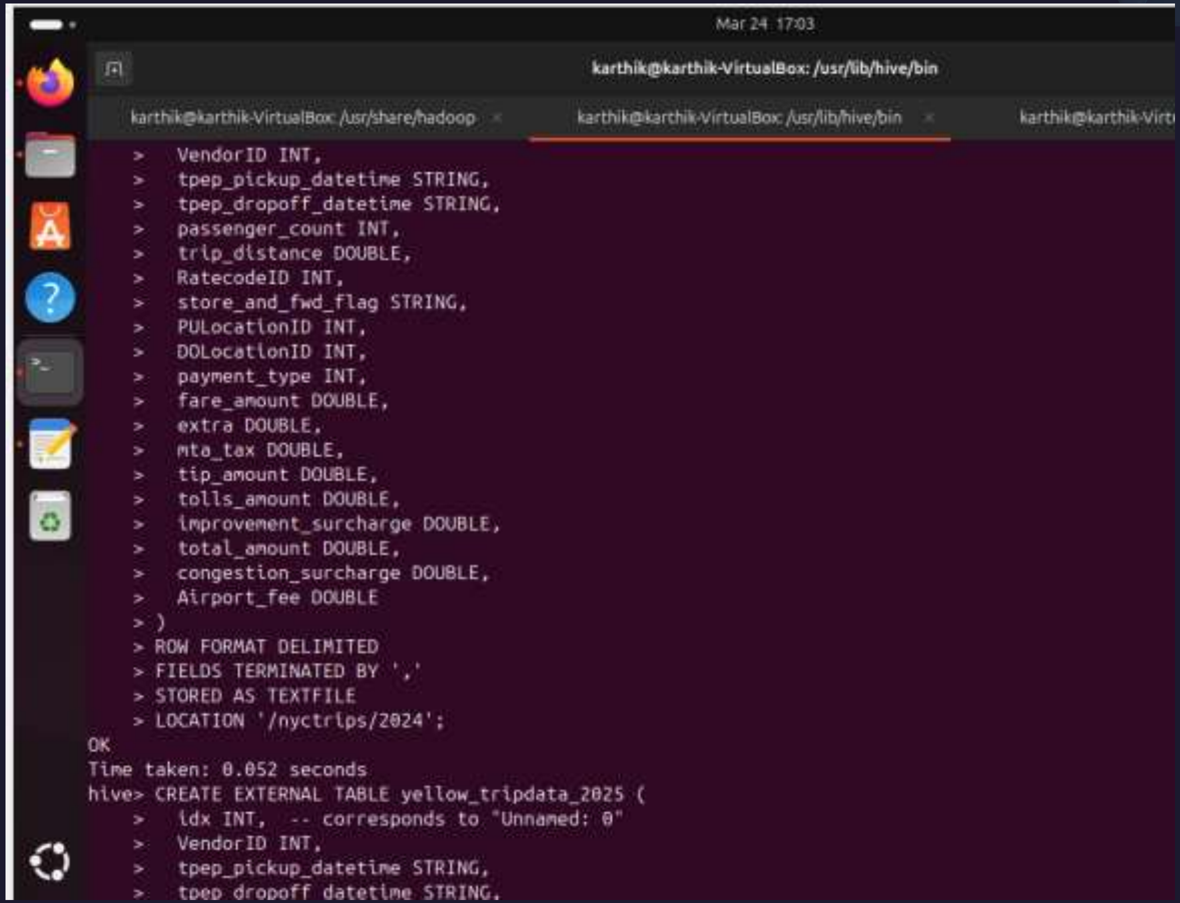


- **Dataset Description:**
  - NYC Yellow Taxi data for January 2024 and January 2025 (CSV files).
- **Storage:**
  - Loaded into Hadoop HDFS (organized by year in separate subdirectories).

- **Methodology**

- **Data Pipeline:**
  - **Step 1:** Data Ingestion into HDFS
  - **Step 2:** Creation of Hive External Tables (one per year)
  - **Step 3:** Execution of Hive SQL queries to answer business questions
  - **Step 4:** Visualization and interpretation of query results

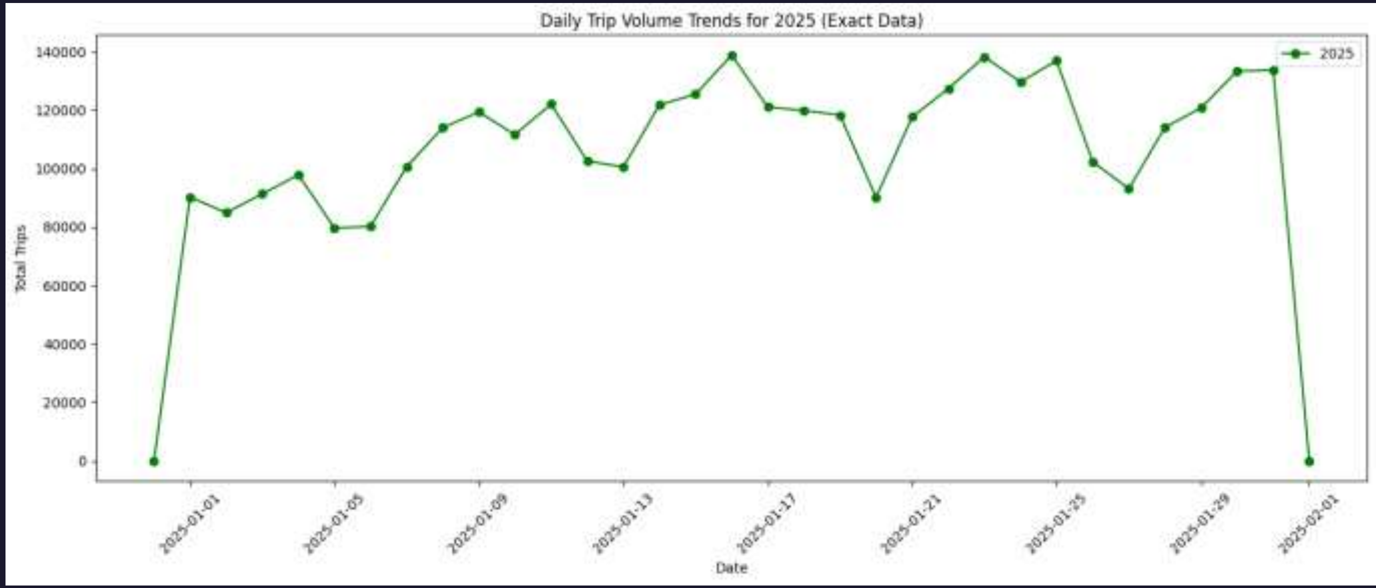
- **High-Level Flow:**
  - Data → HDFS → Hive → Query Results → Visualization

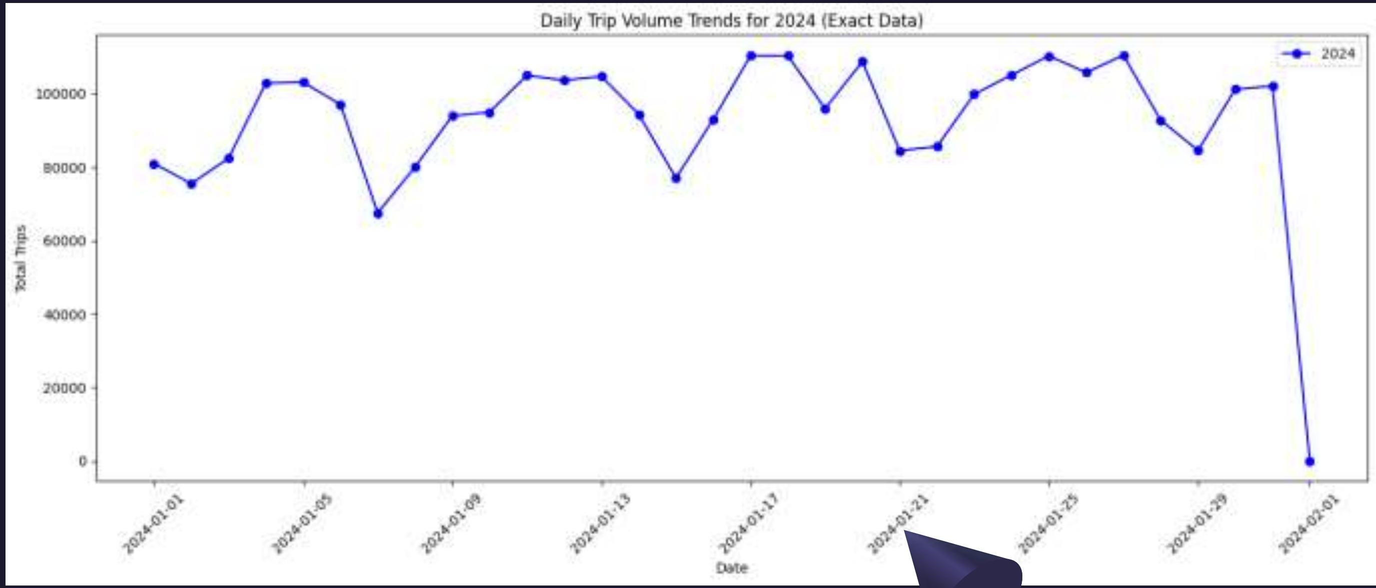


Problem 1: How do daily trip volumes compare between January 2024 and January 2025?

Observed Results:

- The daily trip counts mostly fall within the range of around 80,000 to 110,000 trips for the bulk of January, though there are a few entries with extremely low counts (e.g., early dates like “2002-12-31”, “2009-01-01”, “2023-12-31”, and “2024-02-01”) which appear to be outliers or artifacts.
- When comparing January 2024 with January 2025, the volumes are more in magnitude with some days in one year higher than the other. Cumulatively 2025 are greater than 2024.

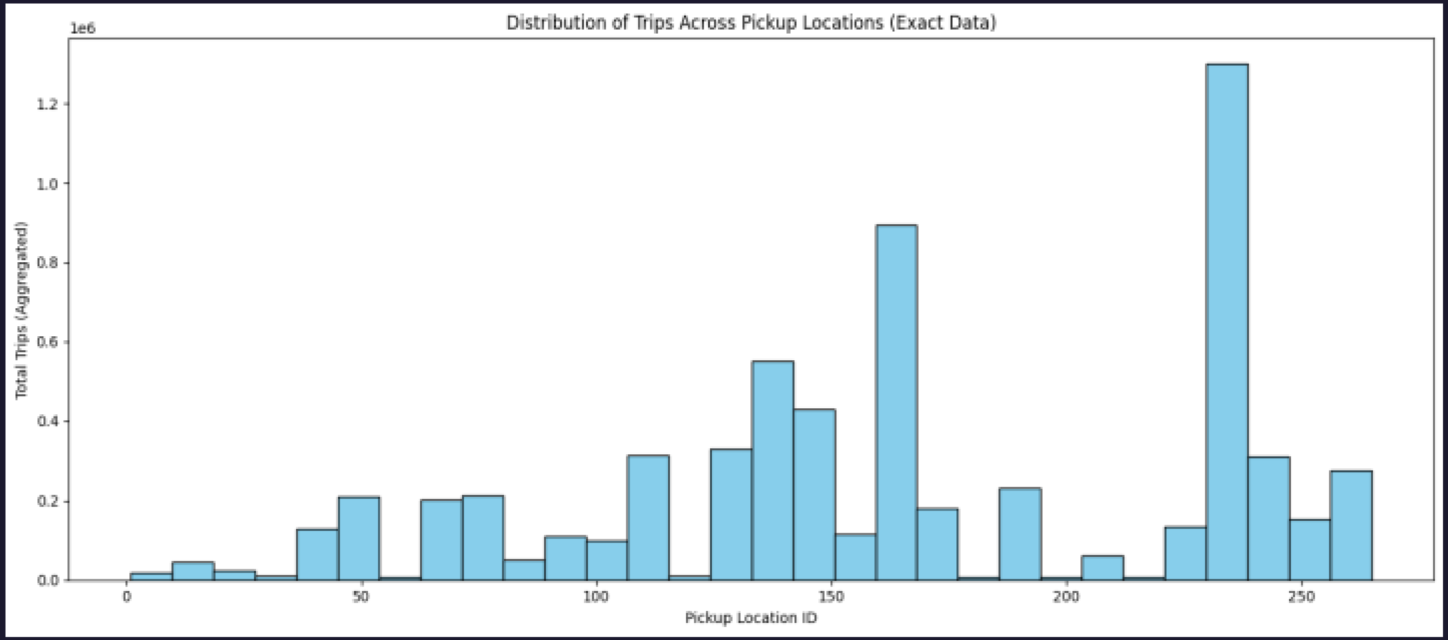


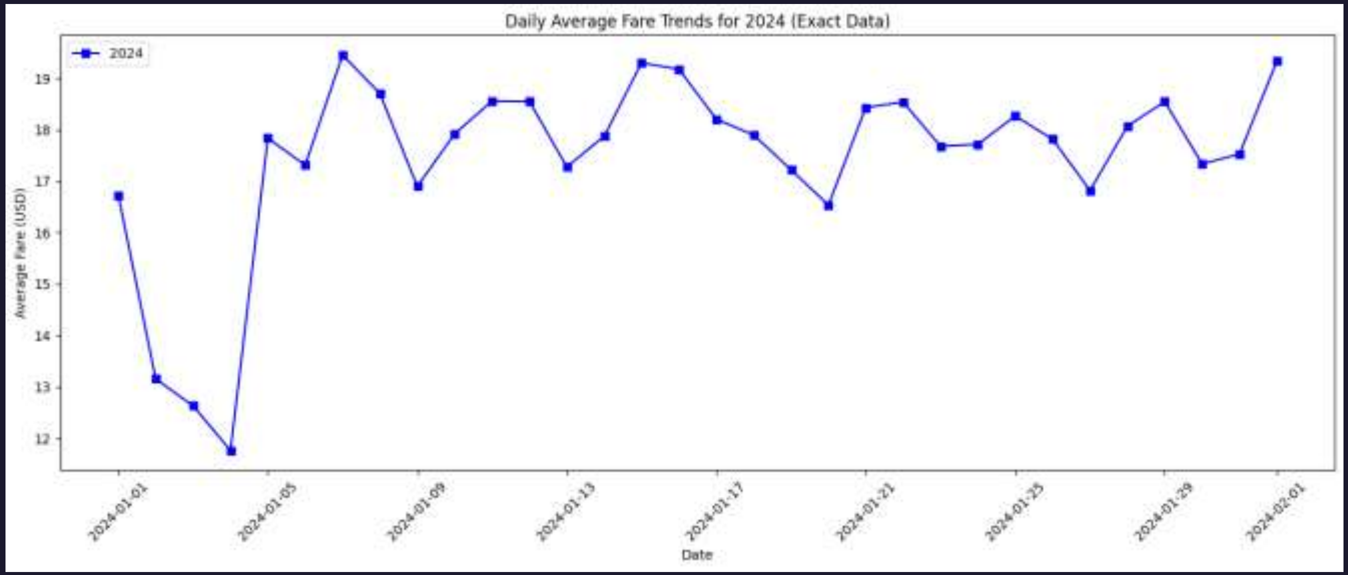


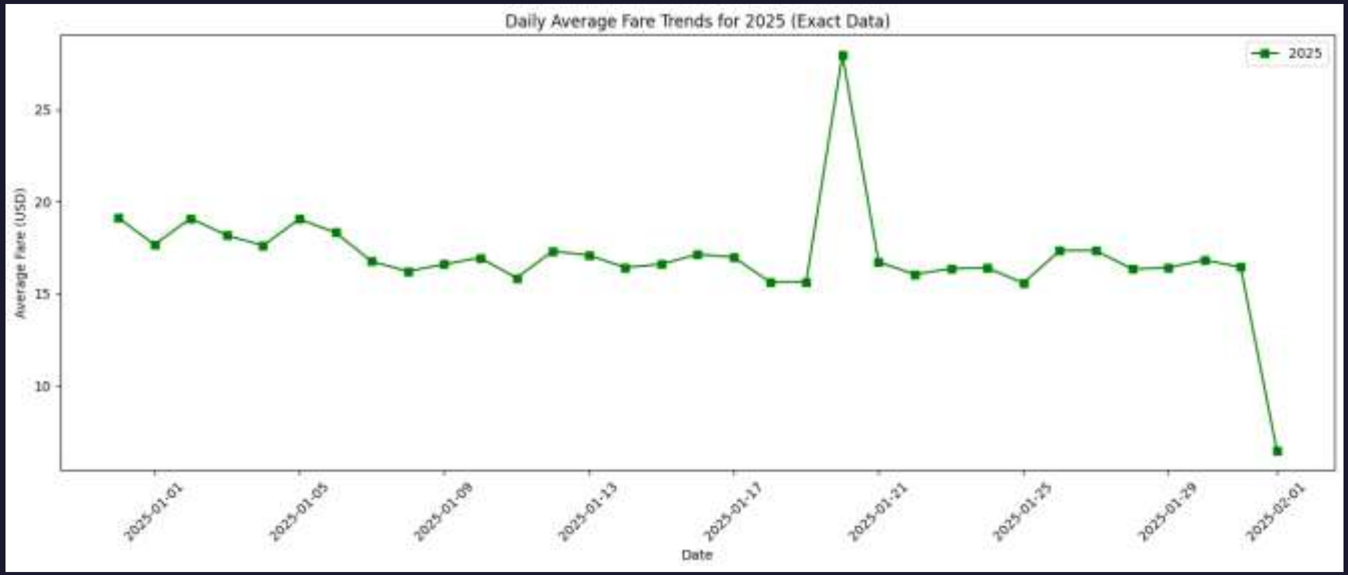
- **Problem 2:** Which pickup locations (PULocationID) have the highest trip counts, and do these change from 2024 to 2025?

- **Observed Results:**

- The results list pickup location IDs (PULocationID) with their corresponding trip counts. Certain location IDs (for example, 161, 237, 236) show particularly high trip counts, indicating that these are major demand hubs.





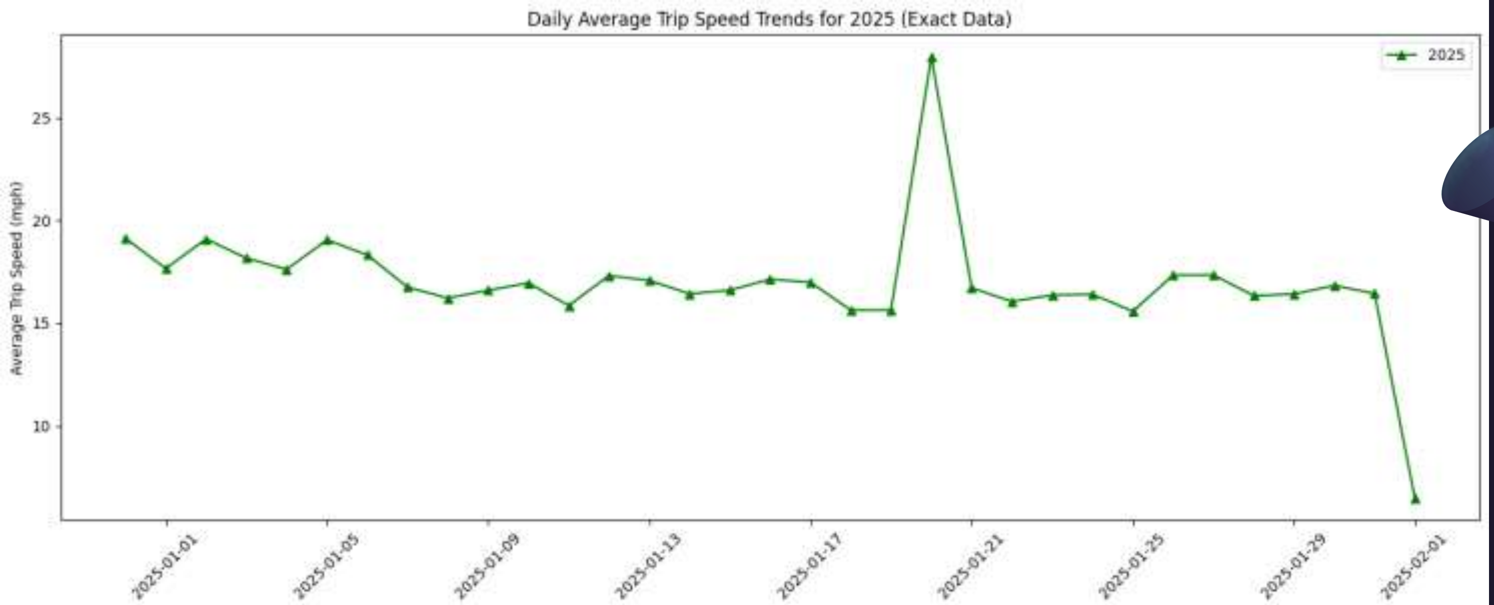


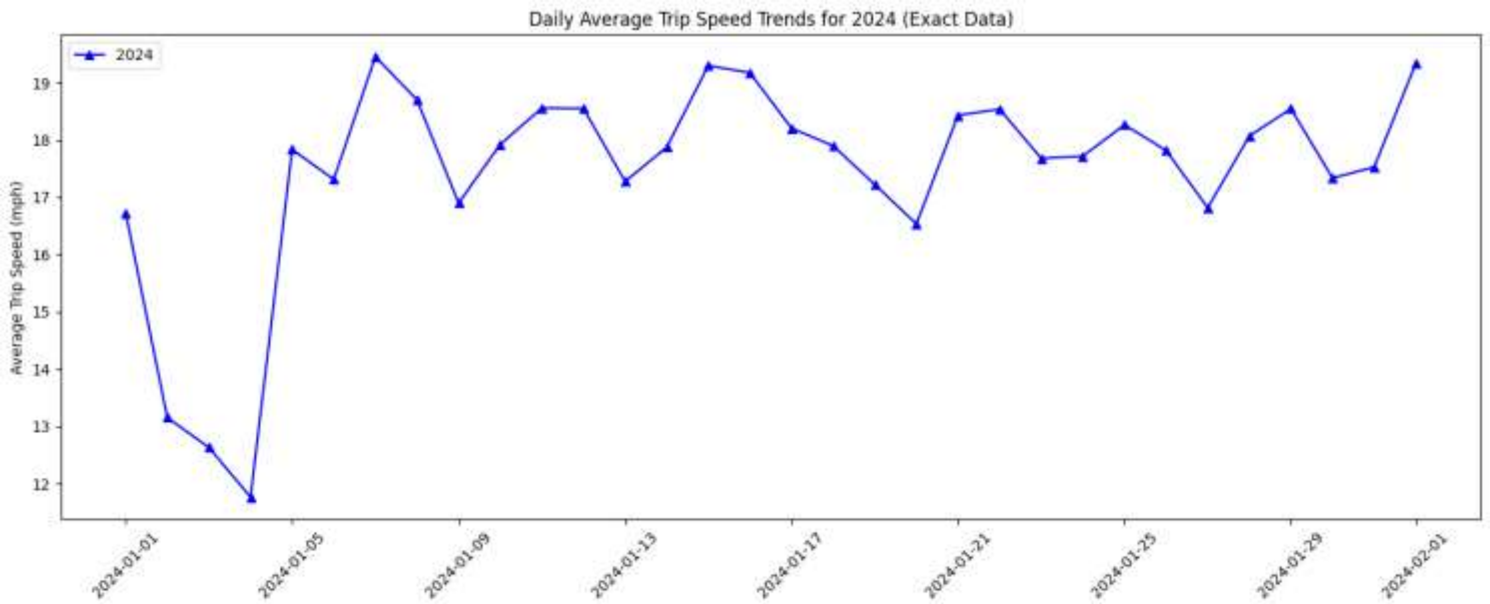
Problem 3: How do average fares per day vary between January 2024 and January 2025?

Observed Results:

- The daily average fares tend to hover around the mid to high teens in US dollars, with most days falling in the 16–19 USD range. However, there are occasional spikes (e.g., a value of 33.33 USD on one day) and very low values (such as 0 or 6.5 USD), which might be outliers.
- Later in the analysis we found that those 2 spikes in the year 2025 are special days which is New Year and Dr. Martin Luther King Jr Day. Every special days there could be a rise the prices.

Problem 4: What are the average trip speeds, and how do they reflect traffic or operational efficiency differences between the two periods?





Thank you